

# Three-Step Equations

Name  
Date  
Course

$$\textcircled{1} \quad 3x + 5x - 6 = 2$$

$$8x - 6 = 2$$

$$6 + 8x - 6 = 2 + 6$$

$$8x = 8$$

$$\frac{8x}{8} = \frac{8}{8}$$

$$\boxed{x = 1}$$

$$\textcircled{2} \quad 4 = 7 - 2y + 8y$$

$$4 = 7 + 6y$$

$$-7 + 4 = 7 + 6y - 7$$

$$-3 = 6y$$

$$\frac{-3}{6} = \frac{6y}{6}$$

$$\boxed{-\frac{1}{2} = y}$$

$$\textcircled{3} \quad -3 + \frac{2z}{3} - \frac{4z}{5} = -4$$

$$-3 + \frac{10z}{15} - \frac{12z}{15} = -4$$

$$3 - 3 - \frac{2z}{15} = -4 + 3$$

$$-\frac{2z}{15} = -1$$

$$(15) \left( -\frac{2z}{15} \right) = -1(15)$$

$$-2z = -15$$

$$\frac{-2z}{-2} = \frac{-15}{-2}$$

$$\boxed{z = \frac{15}{2}}$$

$$\textcircled{4} \quad -5x + x - 7 = -2$$

$$-4x - 7 = -2$$

$$7 - 4x - 7 = -2 + 7$$

$$-4x = 5$$

$$\frac{-4x}{-4} = \frac{5}{-4}$$

$$\boxed{x = -\frac{5}{4}}$$

$$\textcircled{5} \quad -3 = -8y + 10y + 7$$

$$-3 = 2y + 7$$

$$-7 - 3 = 2y + 7 - 7$$

$$-10 = 2y$$

$$\frac{-10}{2} = \frac{2y}{2}$$

$$\boxed{-5 = y}$$

$$\textcircled{6} \quad 7 + \frac{3y}{4} - \frac{7y}{8} = 5$$

$$7 + \frac{6y}{8} - \frac{7y}{8} = 5$$

$$-7 + 7 + \frac{-y}{8} = 5 - 7$$

$$-\frac{y}{8} = -2$$

$$(-8) \frac{y}{8} = -2(-8)$$

$$\boxed{y = 16}$$

$$\textcircled{7} \quad 3 = 8 + \frac{a}{2} + \frac{a}{6}$$

$$3 = 8 + \frac{3a}{6} + \frac{a}{6}$$

$$3 = 8 + \frac{4a}{6}$$

$$3 = 8 + \frac{2a}{3}$$

$$-8 + 3 = 8 + \frac{2a}{3} - 8$$

$$-5 = \frac{2a}{3}$$

$$(3)(-5) = \frac{2a}{3}(3)$$

$$-15 = 2a$$

$$\frac{-15}{2} = \frac{2a}{2}$$

$$\boxed{-\frac{15}{2} = a}$$

$$\textcircled{8} \quad -8b - 2b + 1 = -8$$

$$-10b + 1 = -8$$

$$-1 - 10b + 1 = -8 - 1$$

$$-10b = -9$$

$$\frac{-10b}{-10} = \frac{-9}{-10}$$

$$\boxed{b = \frac{9}{10}}$$

$$\textcircled{9} -5 = 3c - 8c - 2$$

$$-5 = -5c - 2$$

$$2 - 5 = -5c - 2 + 2$$

$$-3 = -5c$$

$$\frac{-3}{-5} = \frac{-5c}{-5}$$

$$\boxed{\frac{3}{5} = c}$$

$$\textcircled{10} -\frac{4x}{9} + \frac{2x}{3} + 3 = -2$$

$$-\frac{4x}{9} + \frac{6x}{9} + 3 = -2$$

$$-3 + \frac{2x}{9} + 3 = -2 - 3$$

$$\frac{2x}{9} = -5$$

$$(9) \frac{2x}{9} = -5(9)$$

$$2x = -45$$

$$\frac{2x}{2} = \frac{-45}{2}$$

$$\boxed{x = -\frac{45}{2}}$$

$$\textcircled{11} -y - y - 5 = -3$$

$$-2y - 5 = -3$$

$$5 - 2y - 5 = -3 + 5$$

$$-2y = 2$$

$$\frac{-2y}{-2} = \frac{2}{-2}$$

$$\boxed{y = -1}$$

$$\textcircled{12} -3 + 4z - 9z = 4$$

$$-3 - 5z = 4$$

$$3 - 3 - 5z = 4 + 3$$

$$-5z = 7$$

$$\frac{-5z}{-5} = \frac{7}{-5}$$

$$\boxed{z = -\frac{7}{5}}$$

$$\textcircled{13} 4x + 2x + 9 = -9$$

$$6x + 9 = -9$$

$$-9 + 6x + 9 = -9 - 9$$

$$6x = -18$$

$$\frac{6x}{6} = \frac{-18}{6}$$

$$\boxed{x = -3}$$

$$\textcircled{14} 8y - 5y - 1 = -3$$

$$3y - 1 = -3$$

$$1 + 3y - 1 = -3 + 1$$

$$3y = -2$$

$$\frac{3y}{3} = \frac{-2}{3}$$

$$\boxed{y = -\frac{2}{3}}$$

$$\textcircled{15} 1 = \frac{2x}{5} + \frac{x}{10} + 6$$

$$1 = \frac{4x}{10} + \frac{x}{10} + 6$$

$$1 = \frac{5x}{10} + 6$$

$$-6 + 1 = \frac{5x}{10} + 6 - 6$$

$$-5 = \frac{5x}{10}$$

$$-5 = \frac{x}{2}$$

$$(2)(-5) = \frac{x}{2}(2)$$

$$\boxed{-10 = x}$$

$$\textcircled{16} -7z + 4z - 5 = 4$$

$$-3z - 5 = 4$$

$$5 - 3z - 5 = 4 + 5$$

$$-3z = 9$$

$$\frac{-3z}{-3} = \frac{9}{-3}$$

$$\boxed{z = -3}$$

$$\textcircled{17} -x + 9x + 5 = 11$$

$$8x + 5 = 11$$

$$-5 + 8x + 5 = 11 - 5$$

$$8x = 6$$

$$\frac{8x}{8} = \frac{6}{8}$$

$$\boxed{x = \frac{3}{4}}$$

$$\textcircled{18} \frac{5y}{6} - \frac{3y}{4} - 7 = -2$$

$$\frac{10y}{12} - \frac{9y}{12} - 7 = -2$$

$$\frac{y}{12} - 7 = -2$$

$$7 + \frac{y}{12} - 7 = -2 + 7$$

$$\frac{y}{12} = 5$$

$$(12) \frac{y}{12} = 5(12)$$

$$\boxed{y = 60}$$

$$\textcircled{19} -2 = -\frac{a}{4} - \frac{a}{2} + 8$$

$$-2 = -\frac{a}{4} - \frac{2a}{4} + 8$$

$$-2 = -\frac{3a}{4} + 8$$

$$-8 - 2 = -\frac{3a}{4} + 8 - 8$$

$$-10 = -\frac{3a}{4}$$

$$(4)(-10) = -\frac{3a}{4}(4)$$

$$-40 = -3a$$

$$\frac{-40}{-3} = \frac{-3a}{-3}$$

$$\boxed{\frac{40}{3} = a}$$

$$\textcircled{20} 7 - 6b + 4b = 6$$

$$7 - 2b = 6$$

$$-7 + 7 - 2b = 6 - 7$$

$$-2b = -1$$

$$\frac{-2b}{-2} = \frac{-1}{-2}$$

$$\boxed{b = \frac{1}{2}}$$

$$\textcircled{21} -4x - 5x + 2 = -5 \quad \textcircled{22} -9y + 11y - 3 = -7$$

$$-9x + 2 = -5$$

$$2y - 3 = -7$$

$$-2 - 9x + 2 = -5 - 2$$

$$3 + 2y - 3 = -7 + 3$$

$$-9x = -7$$

$$2y = -4$$

$$\frac{-9x}{-9} = \frac{-7}{-9}$$

$$\boxed{x = \frac{7}{9}}$$

$$\frac{2y}{2} = \frac{-4}{2}$$

$$\boxed{y = -2}$$

$$\textcircled{23} -\frac{3z}{2} + \frac{z}{3} + 4 = 2$$

$$-\frac{9z}{6} + \frac{2z}{6} + 4 = 2$$

$$-\frac{7z}{6} + 4 = 2$$

$$-4 - \frac{7z}{6} + 4 = 2 - 4$$

$$-\frac{7z}{6} = -2$$

$$(6)\left(-\frac{7z}{6}\right) = -2(6)$$

$$-7z = -12$$

$$\frac{-7z}{-7} = \frac{-12}{-7}$$

$$\boxed{z = \frac{12}{7}}$$

$$\textcircled{24} 4x + 4x - 9 = 1$$

$$8x - 9 = 1$$

$$9 + 8x - 9 = 1 + 9$$

$$8x = 10$$

$$\frac{8x}{8} = \frac{10}{8}$$

$$\boxed{x = \frac{5}{4}}$$

$$\textcircled{25} -7z + 6z + 4 = -8 \quad \textcircled{26} 8a - 9a + 7 = 8$$

$$-z + 4 = -8$$

$$-a + 7 = 8$$

$$\textcolor{red}{-4} - z + 4 = \textcolor{red}{-8} - \textcolor{red}{4} \quad \textcolor{red}{-7} - a + 7 = \textcolor{red}{8} - \textcolor{red}{7}$$

$$-z = -12$$

$$-a = 1$$

$$\frac{-1z}{-1} = \frac{-12}{-1}$$

$$\frac{-1a}{-1} = \frac{1}{-1}$$

$$\boxed{z = 12}$$

$$\boxed{a = -1}$$

$$\textcircled{27} 8c + c - 6 = -6 \quad \textcircled{28} -5x - 6x + 3 = 3$$

$$9c - 6 = -6$$

$$-11x + 3 = 3$$

$$\textcolor{red}{6} + 9c - 6 = \textcolor{red}{-6} + \textcolor{red}{6}$$

$$\textcolor{red}{-3} - 11x + 3 = \textcolor{red}{3} - \textcolor{red}{3}$$

$$9c = 0$$

$$-11x = 0$$

$$\frac{9c}{9} = \frac{0}{9}$$

$$\frac{-11x}{-11} = \frac{0}{-11}$$

$$\boxed{c = 0}$$

$$\boxed{x = 0}$$

$$\textcircled{29} 2a + \frac{3a}{4} - 6 = 2$$

$$\frac{2a}{1} + \frac{3a}{4} - 6 = 2$$

$$\frac{8a}{4} + \frac{3a}{4} - 6 = 2$$

$$\frac{11a}{4} - 6 = 2$$

$$\textcolor{red}{6} + \frac{11a}{4} - 6 = \textcolor{red}{2} + \textcolor{red}{6}$$

$$\frac{11a}{4} = 8$$

$$\textcolor{green}{(4)} \frac{11a}{4} = \textcolor{green}{8(4)}$$

$$11a = 32$$

$$\frac{11a}{11} = \frac{32}{11}$$

$$\boxed{a = \frac{32}{11}}$$

$$\textcircled{31} \frac{7x}{10} - 3x + 1 = 2$$

$$\frac{7x}{10} - \frac{3x}{1} + 1 = 2$$

$$\frac{7x}{10} - \frac{30x}{10} + 1 = 2$$

$$-\frac{23x}{10} + 1 = 2$$

$$\textcolor{red}{-1} - \frac{23x}{10} + 1 = \textcolor{red}{2} - \textcolor{red}{1}$$

$$-\frac{23x}{10} = 1$$

$$\textcolor{green}{(10)} \left( -\frac{23x}{10} \right) = \textcolor{green}{1(10)}$$

$$-23x = 10$$

$$\frac{-23x}{-23} = \frac{10}{-23}$$

$$\boxed{x = -\frac{10}{23}}$$

$$\textcircled{30} -7 = 5 - \frac{x}{2} - x$$

$$-7 = 5 - \frac{x}{2} - \frac{x}{1}$$

$$-7 = 5 - \frac{x}{2} - \frac{2x}{2}$$

$$-7 = 5 - \frac{3x}{2}$$

$$\textcolor{red}{-5} - 7 = \textcolor{red}{5} - \frac{3x}{2} - \textcolor{red}{5}$$

$$-12 = -\frac{3x}{2}$$

$$\textcolor{green}{(2)} (-12) = \textcolor{green}{-} \frac{3x}{2} \textcolor{green}{(2)}$$

$$-24 = -3x$$

$$\frac{-24}{-3} = \frac{-3x}{-3}$$

$$\boxed{8 = x}$$